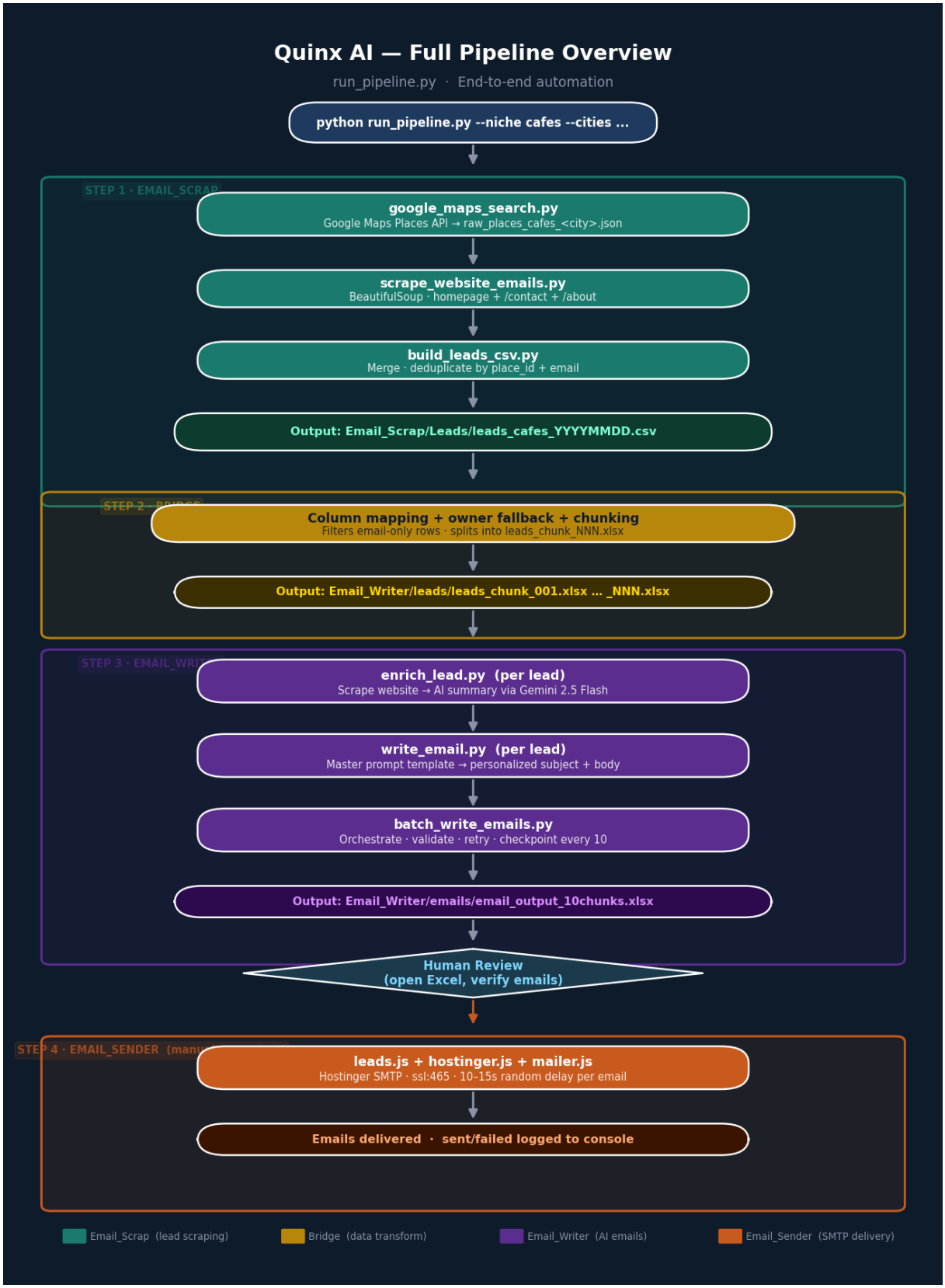
Quinx AI

Automation Architecture

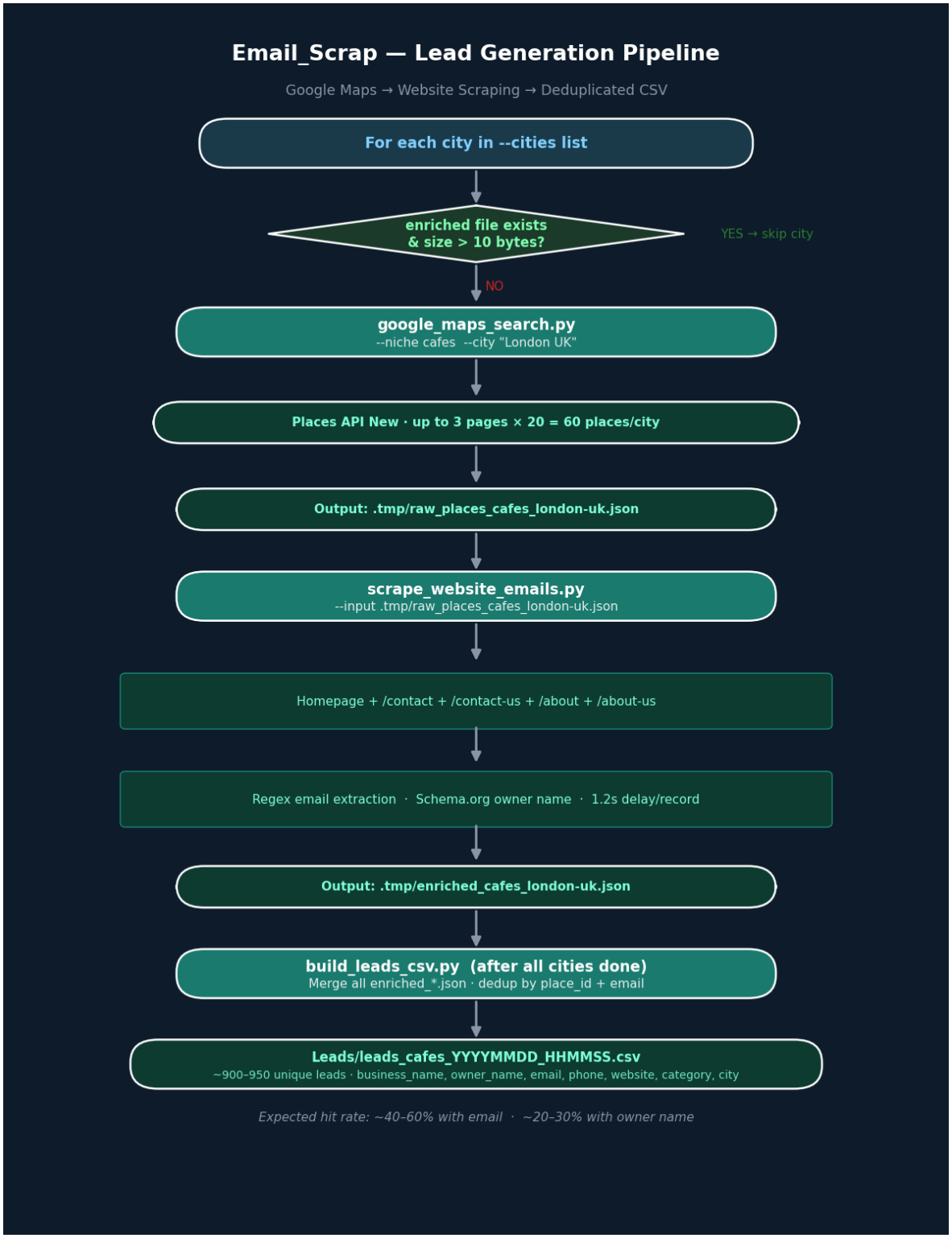
System	End-to-end cold outreach pipeline
Components	Email_Scrap · Email_Writer · Email_Sender
Entry point	python run_pipeline.py --niche <type> --cities <list>
AI Model	Gemini 2.5 Flash via OpenRouter
SMTP	Hostinger · smtp.hostinger.com:465
Output	Personalized cold emails delivered to leads

This document describes the full automation system that scrapes leads, generates personalized cold emails using AI, and delivers them via SMTP — triggered by a single command.

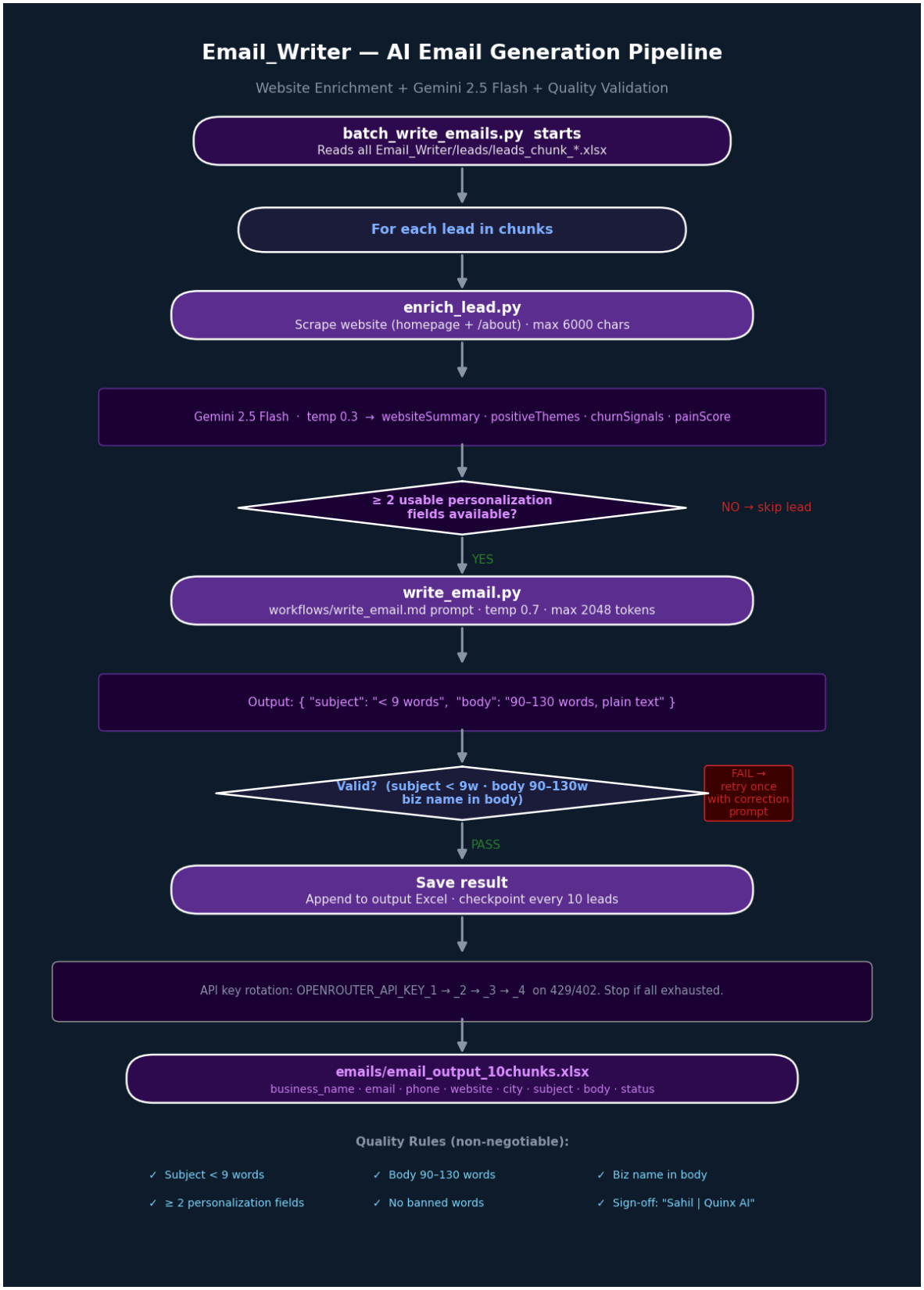
Full Pipeline Overview



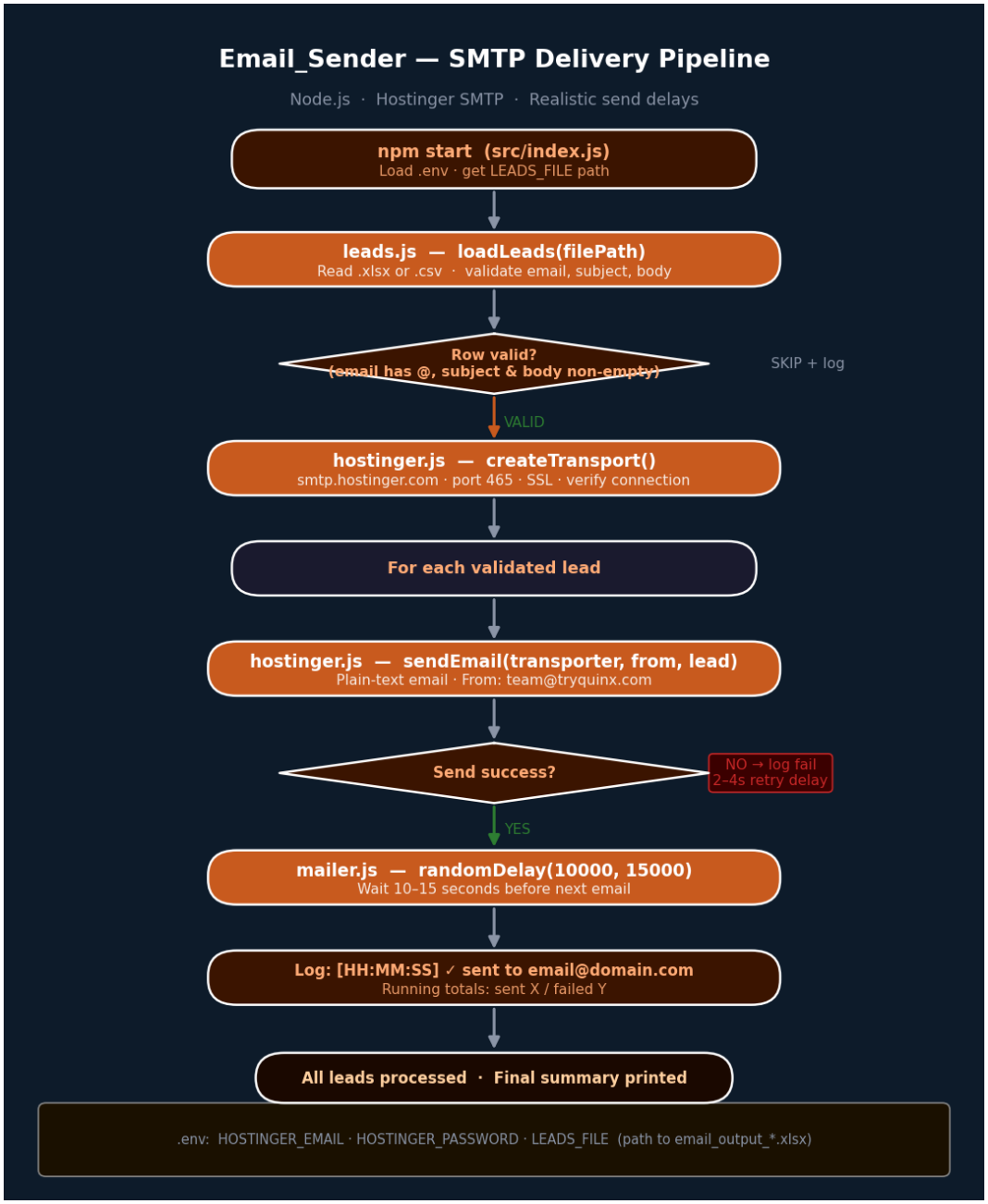
Step 1 · Email_Scrap — Lead Generation



Step 3 · Email_Writer — AI Email Generation



Step 4 · Email_Sender — SMTP Delivery



Data Dictionary

Stage	Fields
Raw (Maps API)	place_id · business_name · address · phone · website · types · city
Enriched (Scraper)	+ email · owner_name
Leads CSV (Email_Scrap)	business_name · owner_name · email · phone · website · category · city · source
Chunk XLSX (Bridge)	business_name · owner_name* · email · phone · website · city · category · niche * fallback: "<Name>'s Team" if blank
Writer Output (Email_Writer)	+ websiteSummary · positiveThemes · churnSignals · painScore · subject · body · status
Sender Input (Email_Sender)	email · subject · body (only these 3 are consumed)

API Keys Reference

Service	Environment Variable	Location	Used by
Google Maps Places API	GOOGLE_MAPS_API_KEY	Email_Scrap/.env	google_maps_search.py
OpenRouter (Gemini 2.5 Flash)	OPENROUTER_API_KEY_1 … _4 (auto-rotate)	Email_Writer/.env	enrich_lead.py write_email.py
Hostinger SMTP	HOSTINGER_EMAIL HOSTINGER_PASSWORD	Email_Sender/.env	hostinger.js

Quick Start

Test run (1 city, fast)

```
python run_pipeline.py --niche "cafes" --cities "London UK"
```

Production (25 cities)

```
python run_pipeline.py --niche "cafes"
```

Custom niche + cities

```
python run_pipeline.py --niche "restaurants" --cities "New York City USA" "Tokyo Japan"
```

After reviewing output Excel

```
cd Email_Sender && npm start
```